Employee Attrition Prediction using Machine Learning (Weka-Based Analysis)

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⭐ **Course: PRP Mini Project**

⭐ **Institute: IIT Jodhpur** **Date:- 13-04-2026**

🟣 1. Corporate Story (Narrative)

TechNova Solutions, a large IT company with 25,000 employees, is experiencing a rise in employee attrition. Project managers are losing skilled employees, HR teams are unable to identify the reasons behind attrition, and leadership needs predictive insights for decision-making.

Employee attrition leads to increased costs, project delays, and inefficient workforce planning. This project uses machine learning techniques to analyze employee data and predict attrition.

🟣 2. Dataset Description

The dataset used is the IBM HR Analytics Attrition Dataset.

- Total Records: 1470
- Target Variable: Attrition (Yes/No)
- Features include Age, Department, JobRole, MonthlyIncome, JobSatisfaction, OverTime, etc.

This dataset is suitable for classification tasks and helps in understanding employee behavior patterns.

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3. Tools Used

- **Google Sheets** → Data cleaning
- **Weka** → Machine learning modeling
- **Flourish** → Data visualization
- **Napkin AI** → Insight generation

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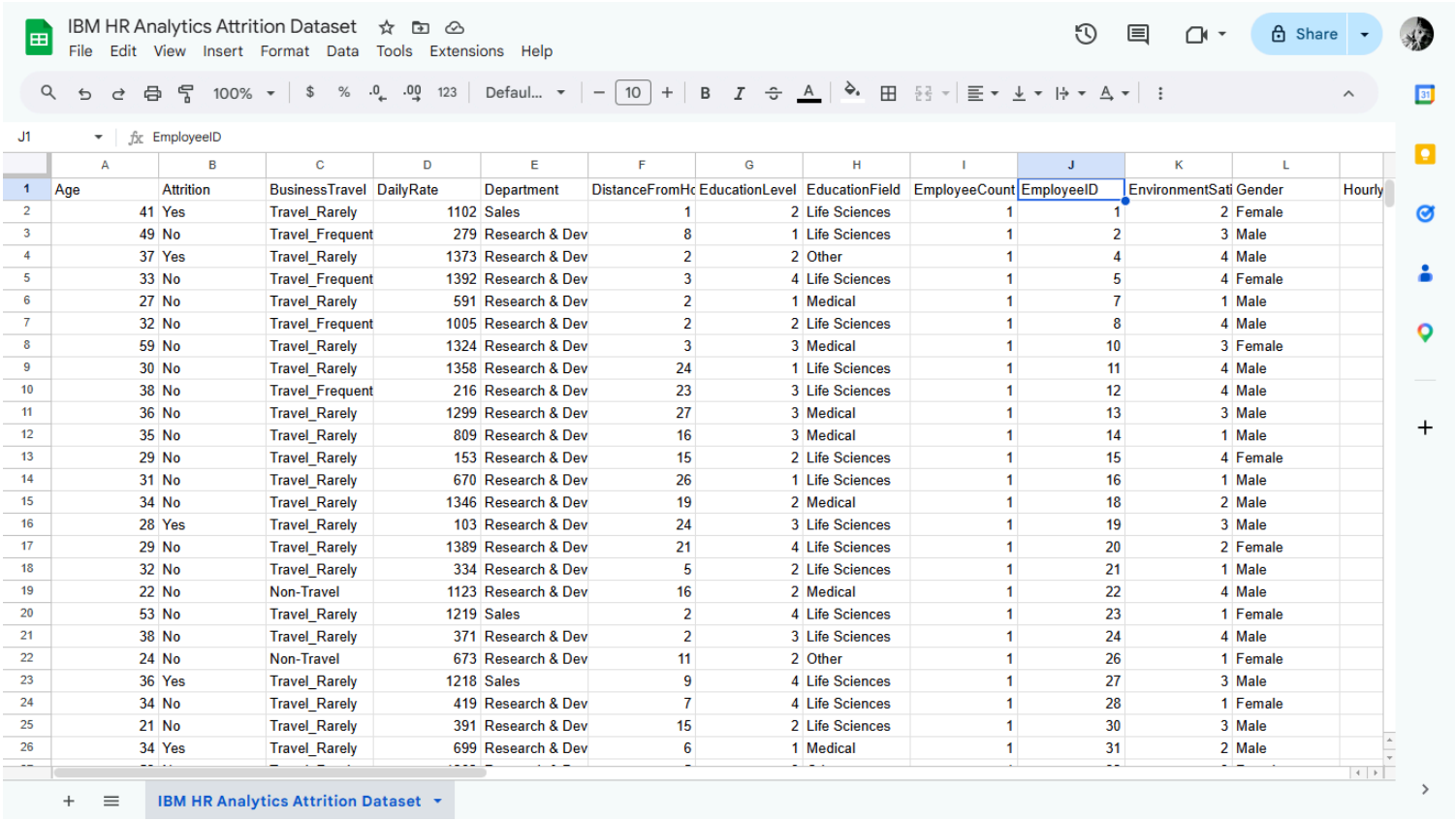
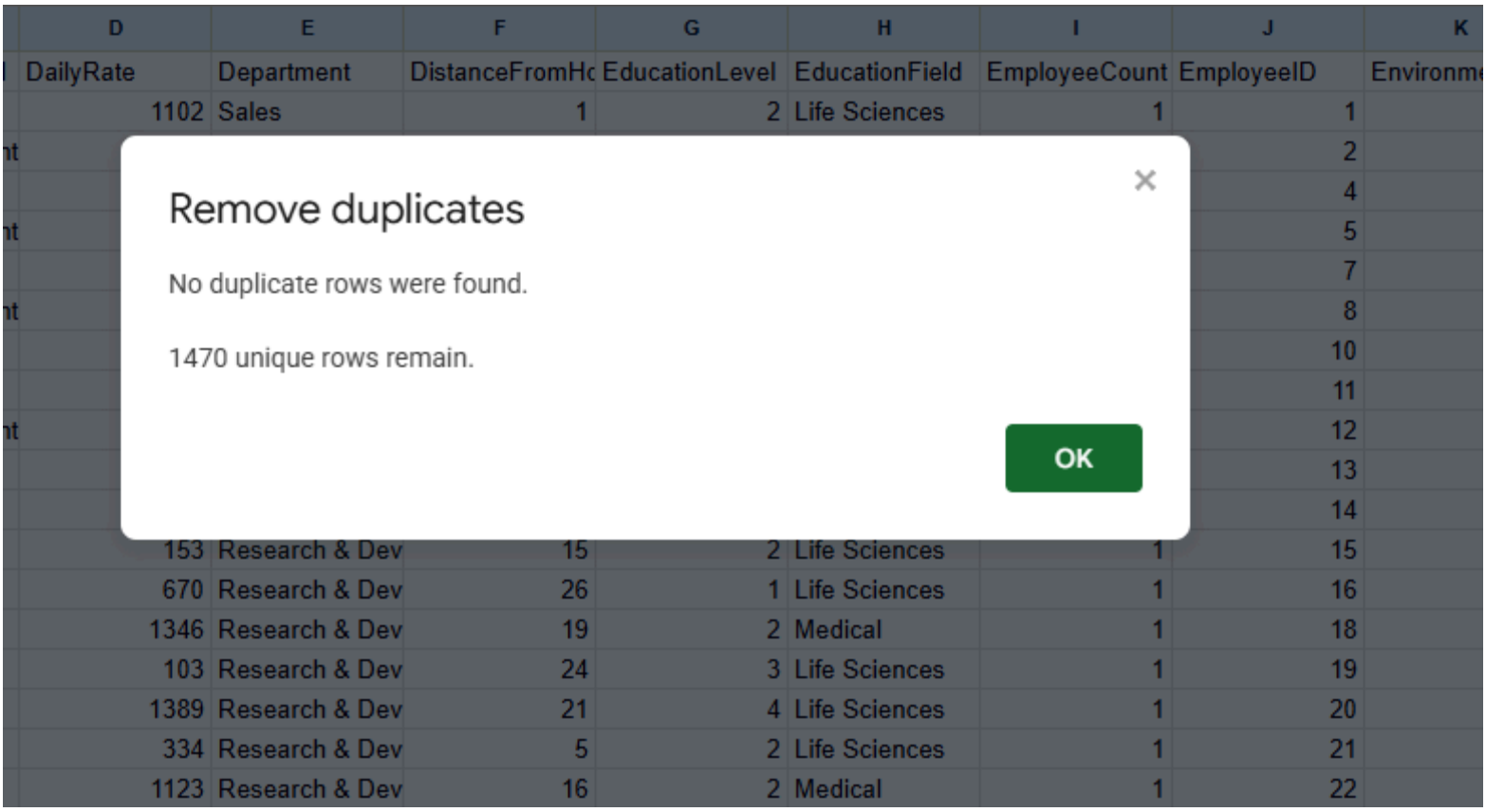
4. Data Cleaning (Google Sheets)

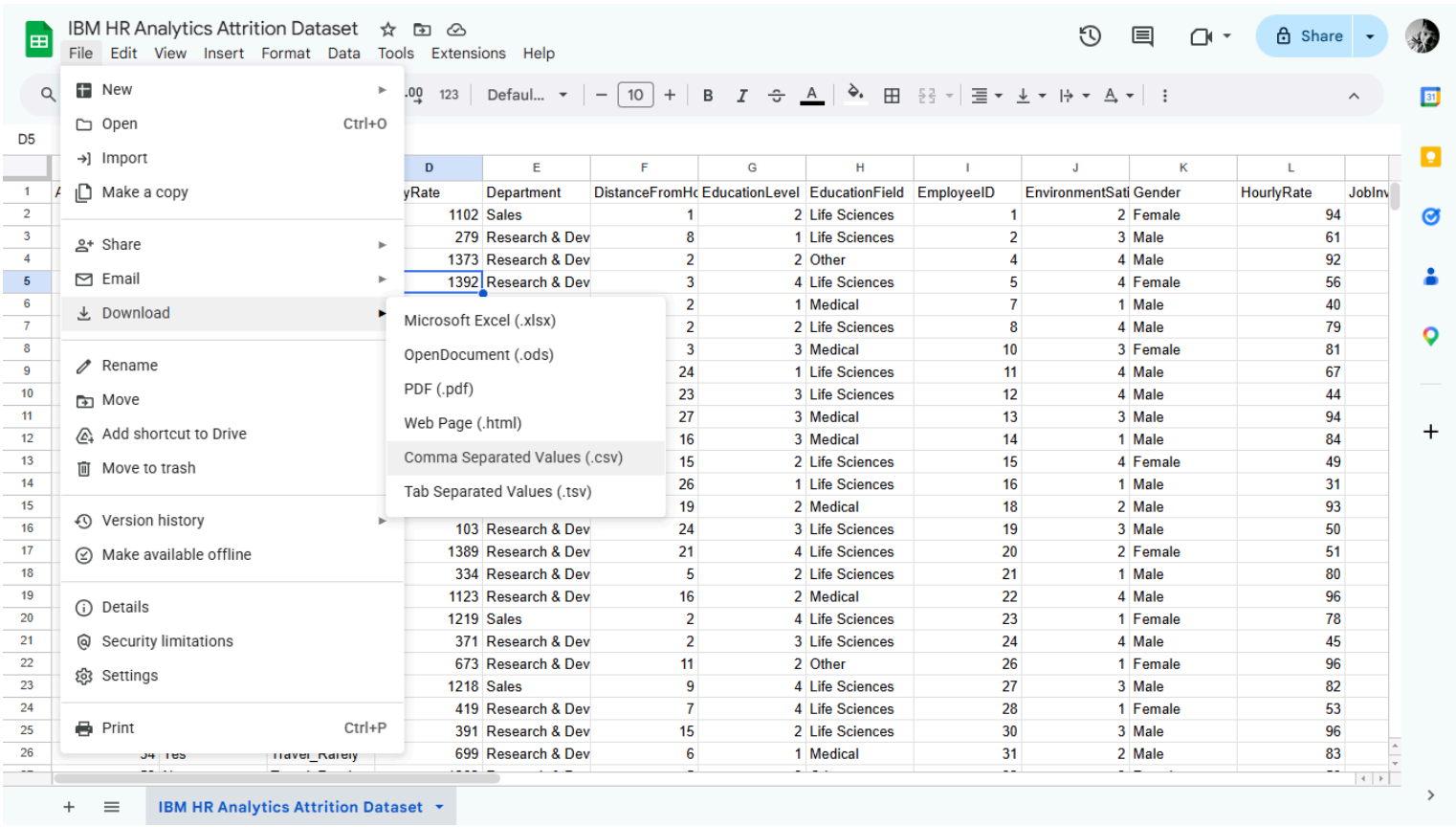
- Dataset imported into Google Sheets
- Duplicates removed
- Columns renamed for clarity
- Data exported as CSV



IBM HR Analytics Attrition Dataset

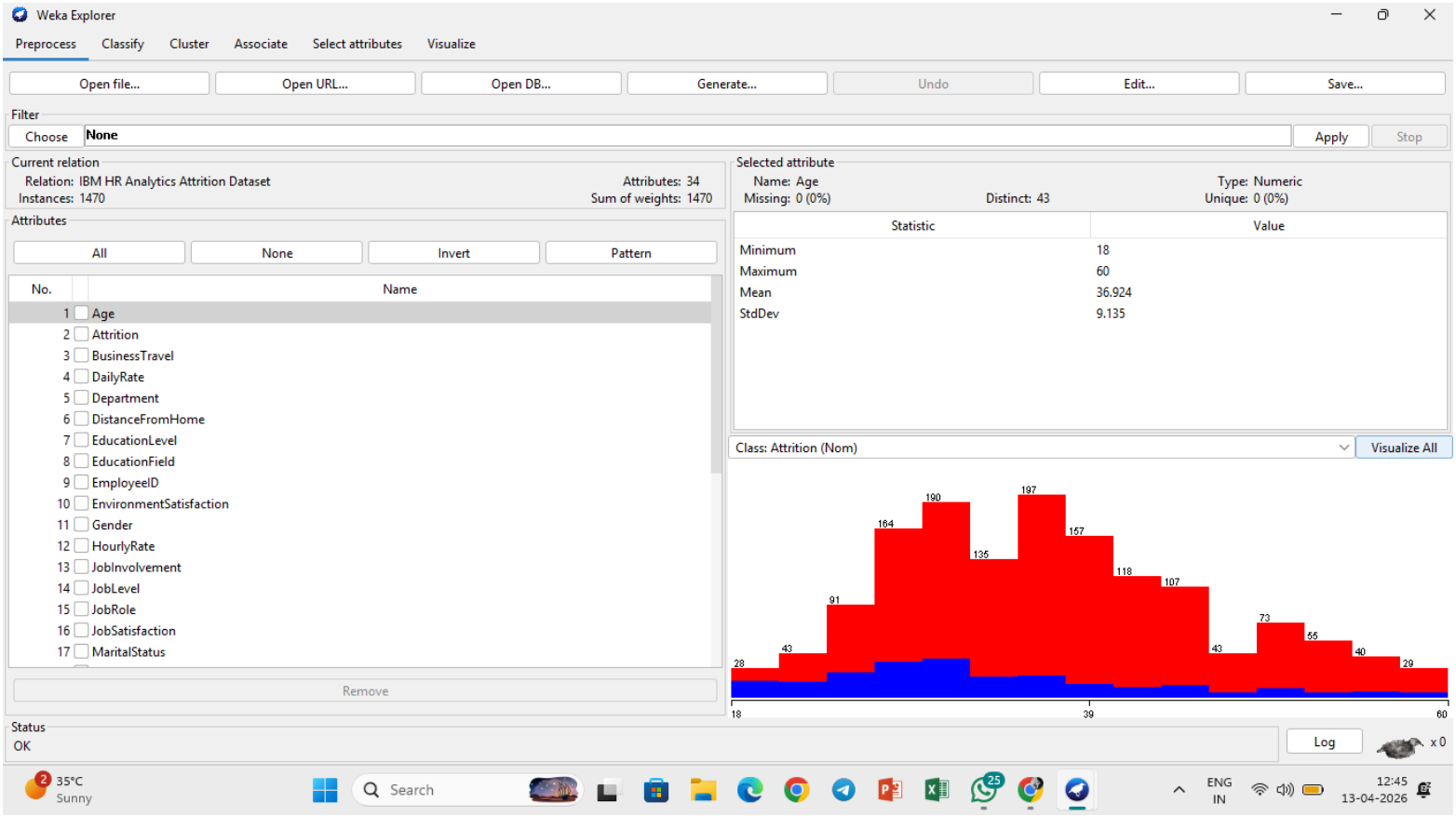
Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHc	EducationLevel	EducationField	EmployeeCount	EmployeeID	EnvironmentSati	Gender	Hourly
41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	2	Female	
49	No	Travel_Frequent	279	Research & Dev	8	1	Life Sciences	1	2	3	Male	
37	Yes	Travel_Rarely	1373	Research & Dev	2	2	Other	1	4	4	Male	
33	No	Travel_Frequent	1392	Research & Dev	3	4	Life Sciences	1	5	4	Female	
27	No	Travel_Rarely	591	Research & Dev	2	1	Medical	1	7	1	Male	
32	No	Travel_Frequent	1005	Research & Dev	2	2	Life Sciences	1	8	4	Male	
59	No	Travel_Rarely	1324	Research & Dev	3	3	Medical	1	10	3	Female	
30	No	Travel_Rarely	1358	Research & Dev	24	1	Life Sciences	1	11	4	Male	
38	No	Travel_Frequent	216	Research & Dev	23	3	Life Sciences	1	12	4	Male	
36	No	Travel_Rarely	1299	Research & Dev	27	3	Medical	1	13	3	Male	
35	No	Travel_Rarely	809	Research & Dev	16	3	Medical	1	14	1	Male	
29	No	Travel_Rarely	153	Research & Dev	15	2	Life Sciences	1	15	4	Female	
31	No	Travel_Rarely	670	Research & Dev	26	1	Life Sciences	1	16	1	Male	
34	No	Travel_Rarely	1346	Research & Dev	19	2	Medical	1	18	2	Male	
28	Yes	Travel_Rarely	103	Research & Dev	24	3	Life Sciences	1	19	3	Male	
29	No	Travel_Rarely	1389	Research & Dev	21	4	Life Sciences	1	20	2	Female	
32	No	Travel_Rarely	334	Research & Dev	5	2	Life Sciences	1	21	1	Male	
22	No	Non-Travel	1123	Research & Dev	16	2	Medical	1	22	4	Male	
53	No	Travel_Rarely	1219	Sales	2	4	Life Sciences	1	23	1	Female	
38	No	Travel_Rarely	371	Research & Dev	2	3	Life Sciences	1	24	4	Male	
24	No	Non-Travel	673	Research & Dev	11	2	Other	1	26	1	Female	
36	Yes	Travel_Rarely	1218	Sales	9	4	Life Sciences	1	27	3	Male	
34	No	Travel_Rarely	419	Research & Dev	7	4	Life Sciences	1	28	1	Female	
21	No	Travel_Rarely	391	Research & Dev	15	2	Life Sciences	1	30	3	Male	
34	Yes	Travel_Rarely	699	Research & Dev	6	1	Medical	1	31	2	Male	


Remove duplicates
No duplicate rows were found.
1470 unique rows remain.
OK


IBM HR Analytics Attrition Dataset
File Edit View Insert Format Data Tools Extensions Help
Share
New
Open Ctrl+O
Import
Make a copy
Share
Email
Download
Rename
Move
Add shortcut to Drive
Move to trash
Version history
Make available offline
Details
Security limitations
Settings
Print Ctrl+P
Microsoft Excel (.xlsx)
OpenDocument (.ods)
PDF (.pdf)
Web Page (.html)
Comma Separated Values (.csv)
Tab Separated Values (.tsv)
IBM HR Analytics Attrition Dataset

5.Data Preparation in Weka

- Dataset loaded in Weka Explorer
- Target variable set as Attrition



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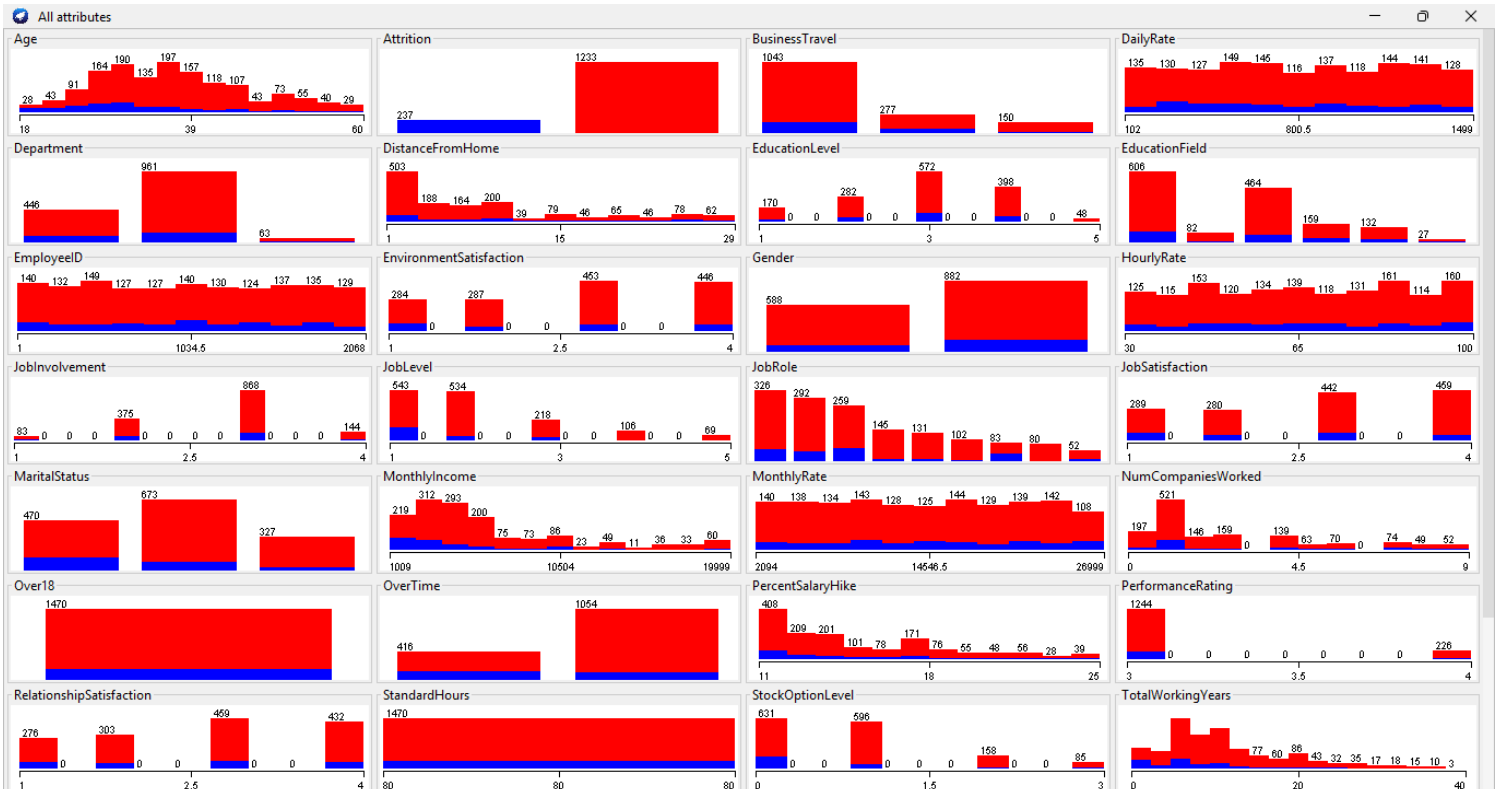
6. Data Visualization (Weka)

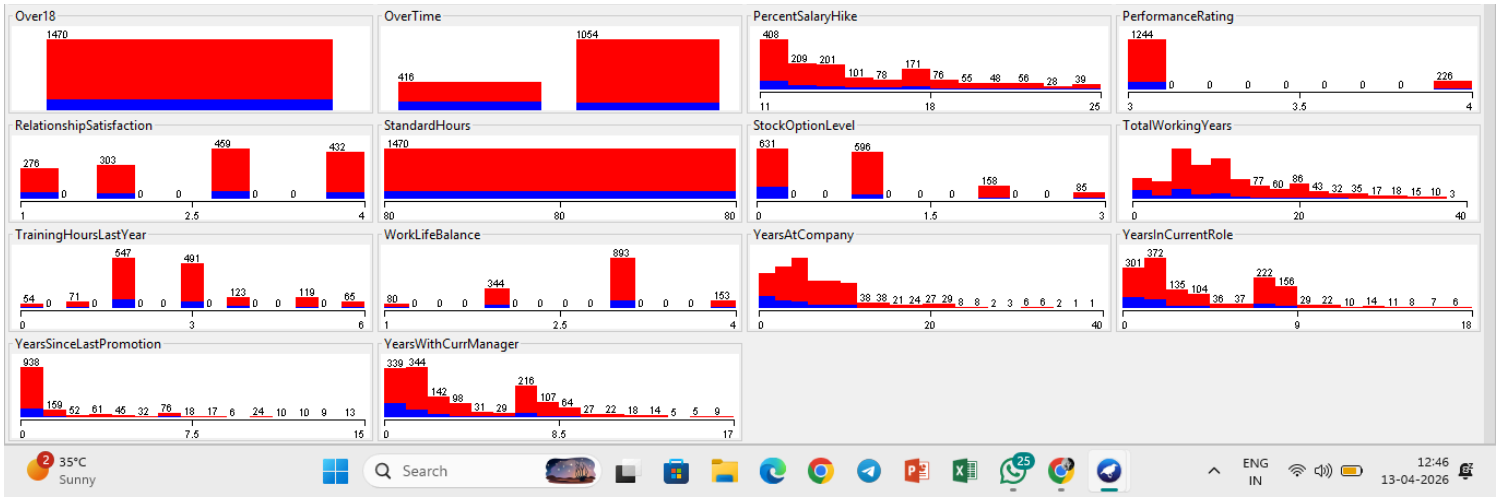
- Attribute relationships explored
- Observed patterns between:
  - Job Satisfaction & Attrition
  - OverTime & Attrition

👉 Visualization of All Attributes 🔍

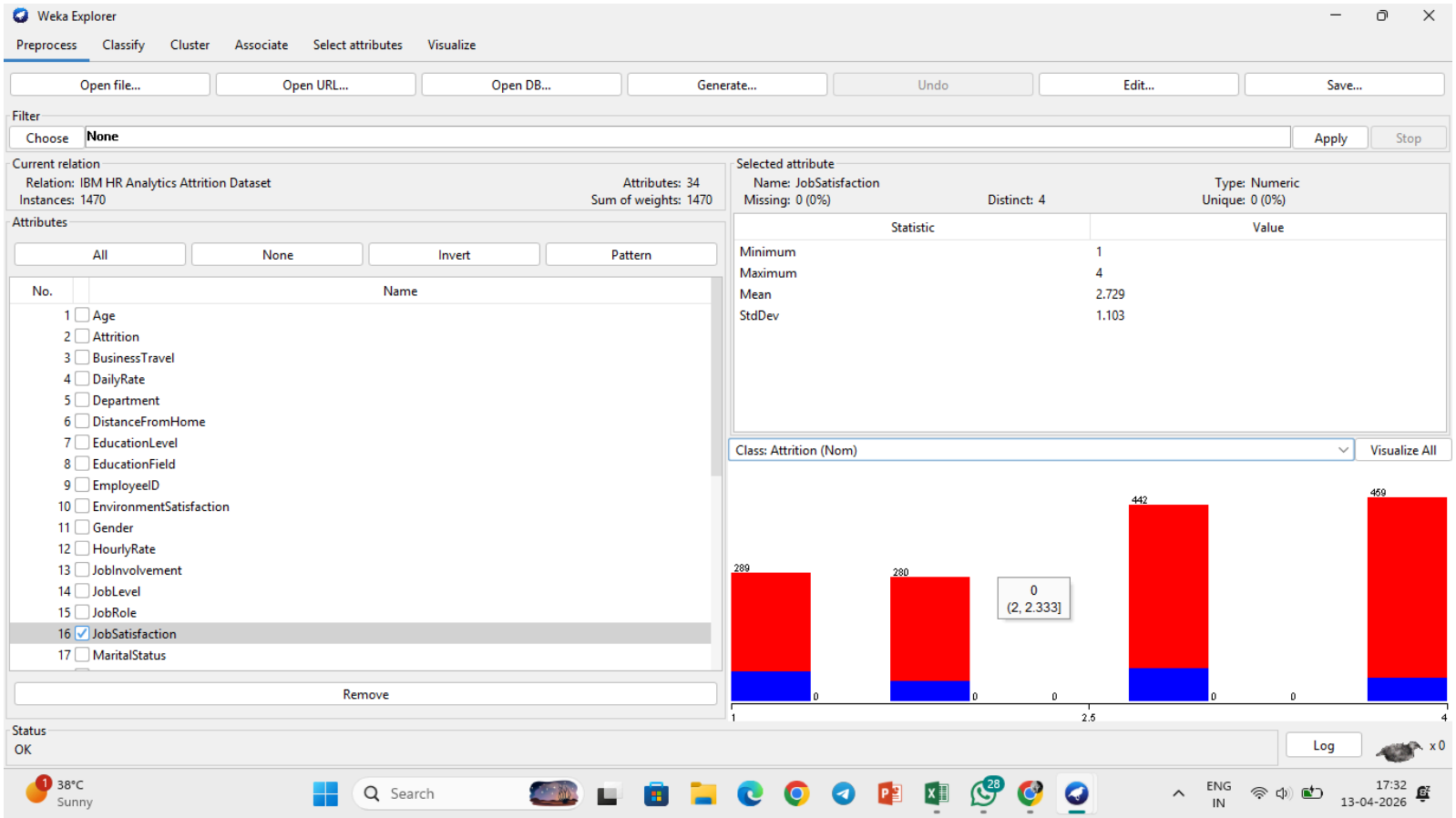
Different attributes were visualized in Weka to understand patterns in the data.t was observed that employees with low job satisfaction are more likely to leave the company. Also, employees who work overtime have a higher chance of attrition.

A histogram was used to understand the distribution of age in the dataset.



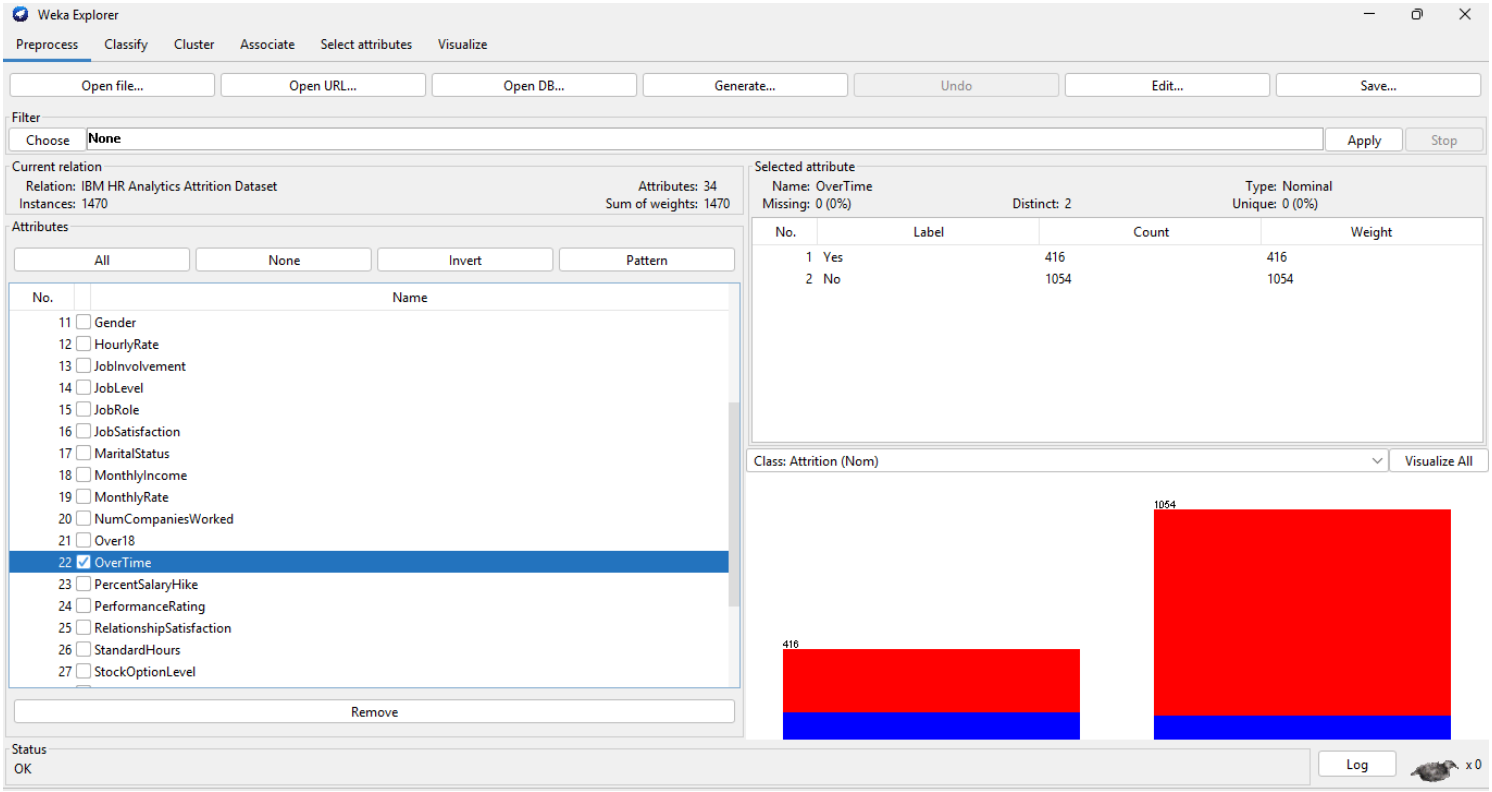


6.1 Attrition vs Job Satisfaction:- It was observed that employees with lower job satisfaction are more likely to leave the company.



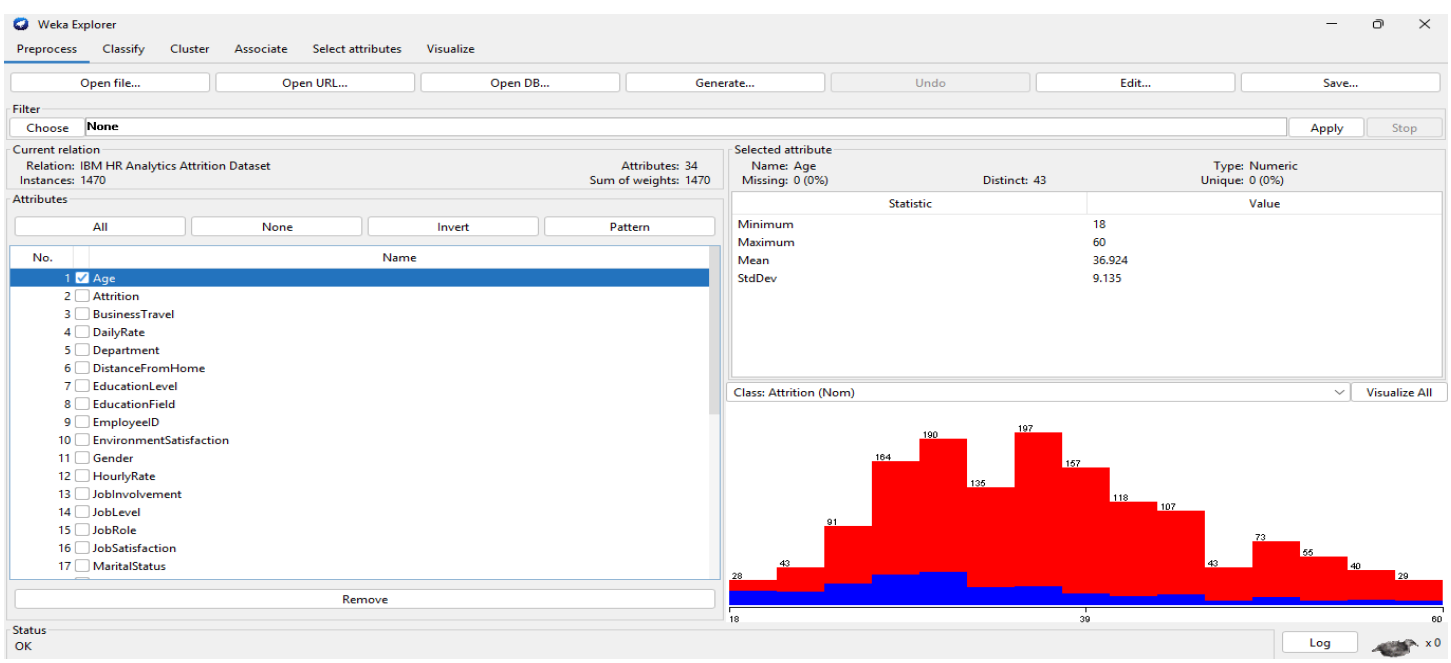
6.2 Attrition vs OverTime

Employees who work overtime have a higher chance of leaving the company.



6.3 Age Distribution (Histogram)

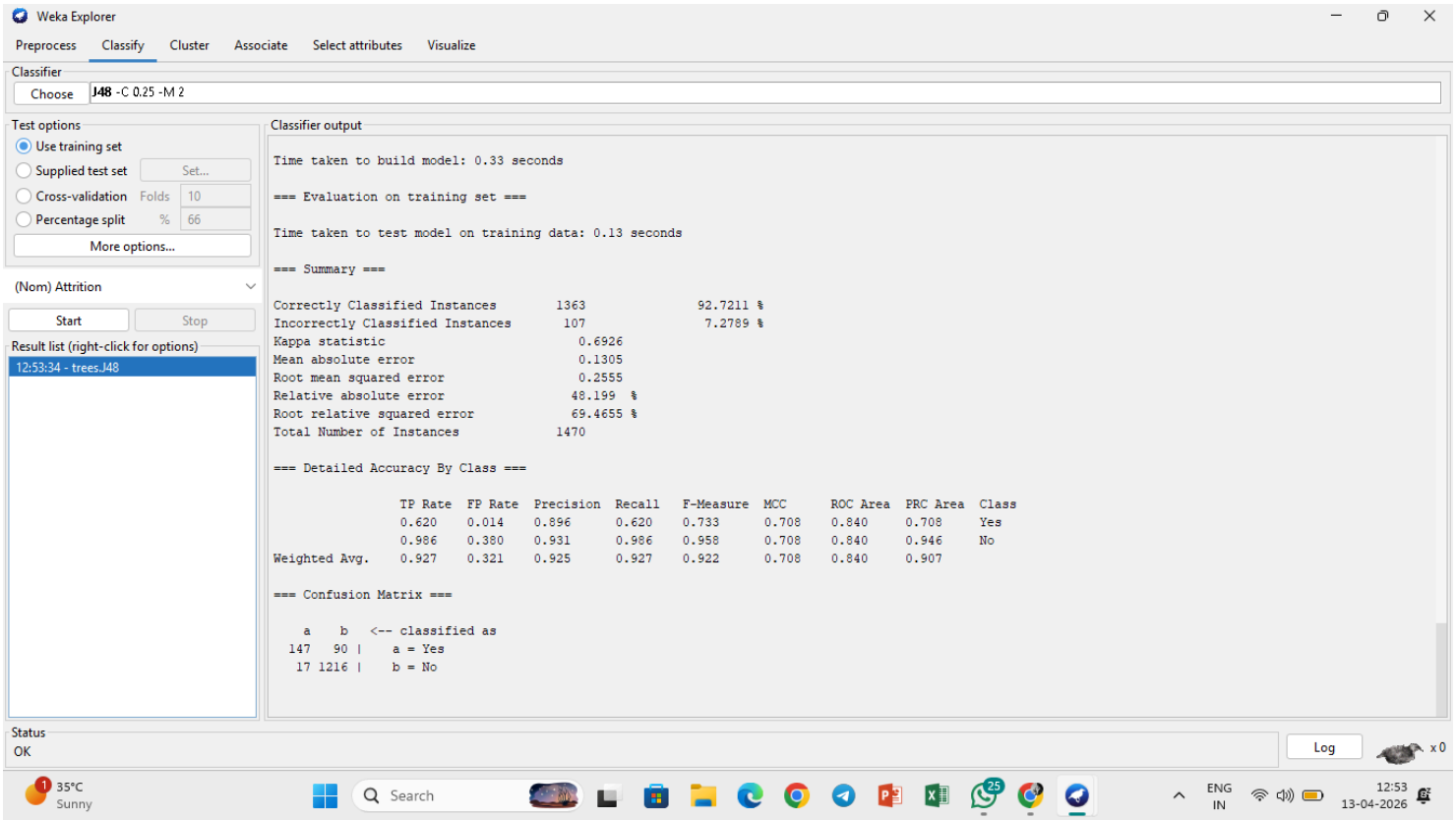
The histogram shows that most employees fall in the mid-age range.



7. Baseline Model (J48 Decision Tree)

- Algorithm: J48
- Accuracy: **92.72%**

The model performed well but showed some errors in predicting employees who leave.



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8. Overfitting Analysis

- Modified model (unpruned J48)
- Training Accuracy: **100%**
- Cross-validation Accuracy: **77.68%**

This shows the model memorized training data but failed to generalize.

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Classifier

Choose: J48 -U -M 1

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds 10
- Percentage split — % 66

More options...

(Nom) Attrition

Start | Stop

Result list (right-click for options)

- 12:53:34 - trees.J48
- 13:09:58 - trees.J48

Classifier output

```
Time taken to build model: 0.16 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.07 seconds

=== Summary ===

Correctly Classified Instances        1470               100      %
Incorrectly Classified Instances         0                 0      %
Kappa statistic                          1
Mean absolute error                      0
Root mean squared error                  0
Relative absolute error                  0      %
Root relative squared error              0      %
Total Number of Instances             1470

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000     Yes
                 1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000     No
Weighted Avg.    1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000

=== Confusion Matrix ===

    a    b   <-- classified as
  237    0 |    a = Yes
    0 1233 |    b = No
```

Status: OK

Log

Hot days ahead 35°C

Search

ENG IN 13:10 13-04-2026

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Classifier

Choose: J48 -U -M 1

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds 10
- Percentage split — % 66

More options...

(Nom) Attrition

Start | Stop

Result list (right-click for options)

- 12:53:34 - trees.J48
- 13:09:58 - trees.J48
- 13:11:06 - trees.J48

Classifier output

```
Size of the tree :      452


Time taken to build model: 0.14 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        1142               77.6871 %
Incorrectly Classified Instances       328               22.3129 %
Kappa statistic                          0.1942
Mean absolute error                      0.2235
Root mean squared error                  0.4707
Relative absolute error                 82.519  %
Root relative squared error            127.9942 %
Total Number of Instances             1470

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.338    0.139    0.319      0.338    0.328      0.194    0.601     0.216     Yes
                 0.861    0.662    0.871      0.861    0.866      0.194    0.601     0.867     No
Weighted Avg.    0.777    0.578    0.782      0.777    0.779      0.194    0.601     0.762

=== Confusion Matrix ===

    a    b   <-- classified as
   80  157 |    a = Yes
  171 1062 |    b = No
```

Status: OK

Log

Hot days ahead 35°C

Search

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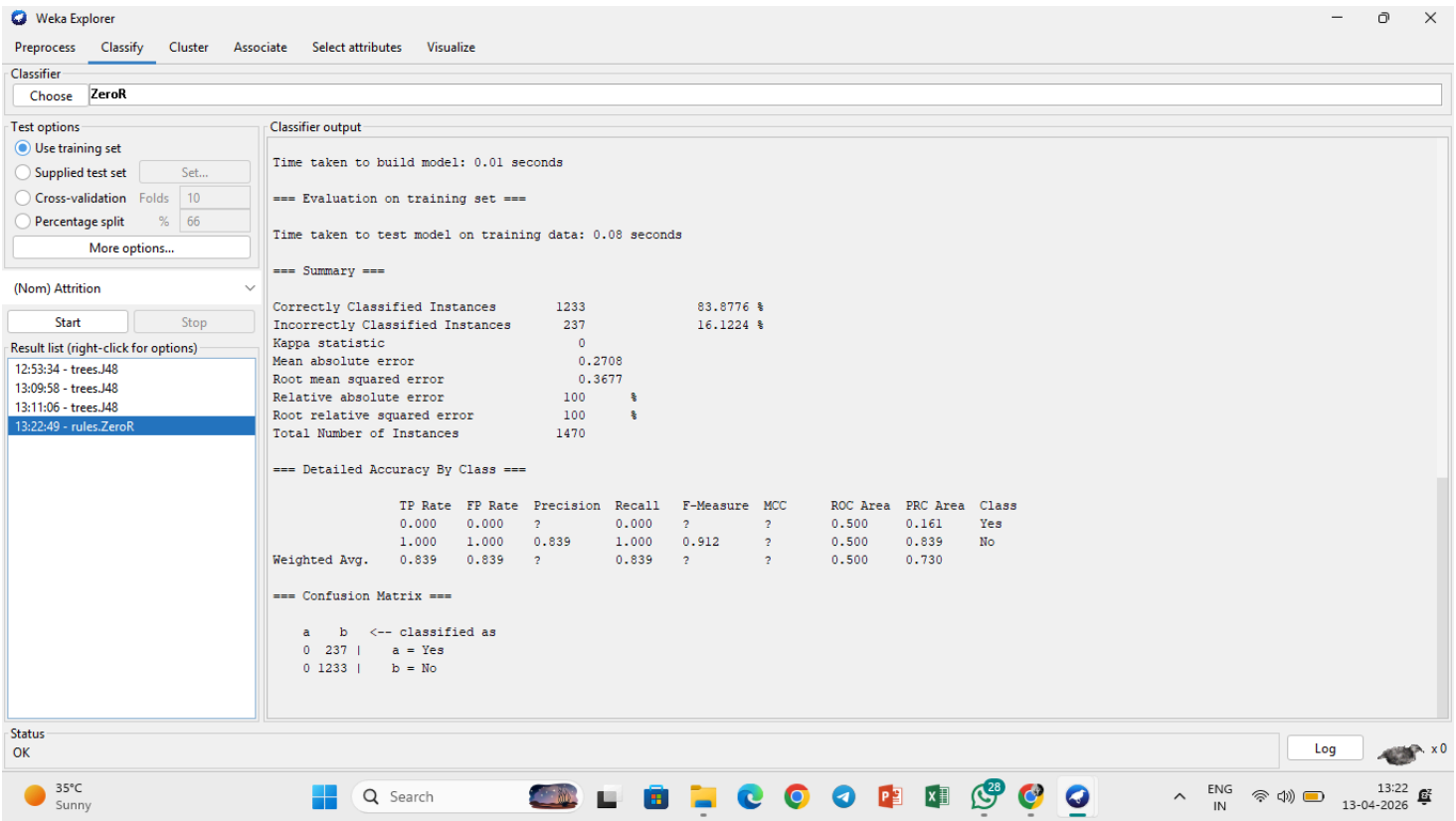
9. Underfitting Analysis

- Model used: ZeroR
- Accuracy: **83.87%**

The model predicted only the majority class and failed to learn patterns.



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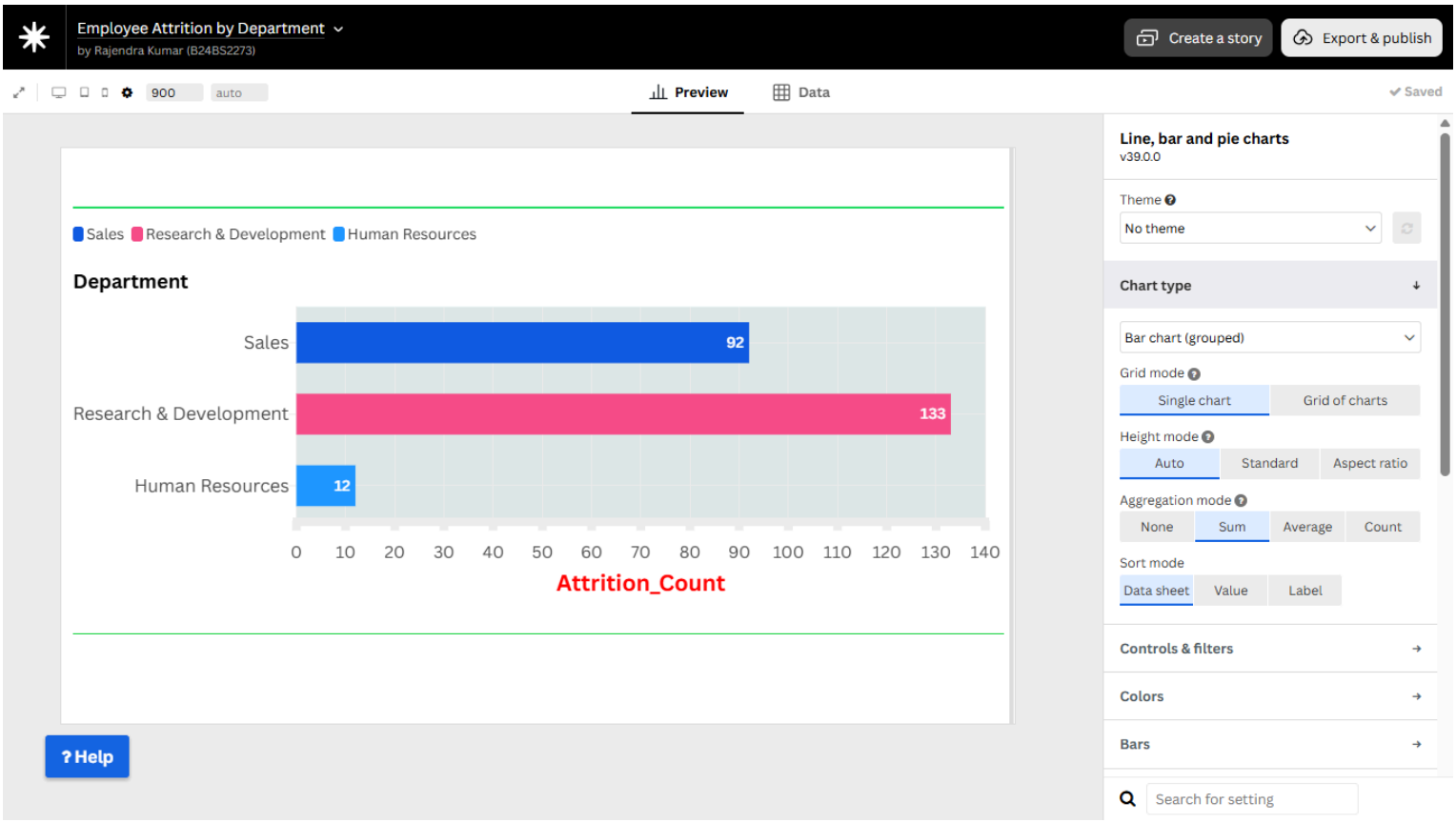
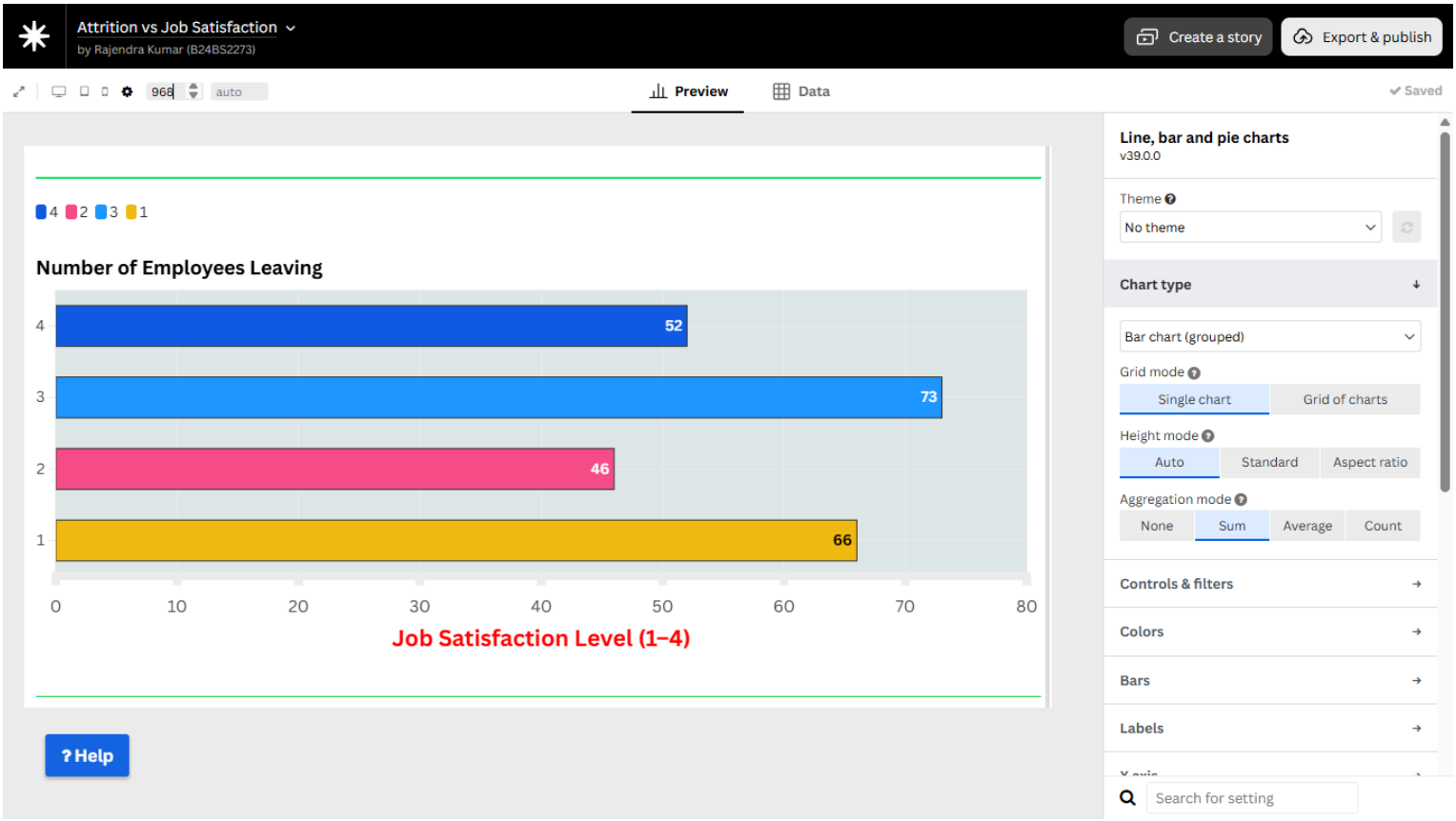
10. Flourish Visualizations

- Attrition by Department
- Attrition vs Job Satisfaction

Insights:

- R&D department has highest attrition
- Low job satisfaction leads to higher attrition


Employee Attrition by Department
by Rajendra Kumar (B24BS2273)
Create a story
Export & publish
900
auto
Preview
Data
Saved
Sales
Research & Development
Human Resources
Department
Sales
Research & Development
Human Resources
92
133
12
0 10 20 30 40 50 60 70 80 90 100 110 120 130 140
Attrition_Count
Help
Line, bar and pie charts
v39.0.0
Theme
No theme
Chart type
Bar chart (grouped)
Grid mode
Single chart
Grid of charts
Height mode
Auto
Standard
Aspect ratio
Aggregation mode
None
Sum
Average
Count
Sort mode
Data sheet
Value
Label
Controls & filters
Colors
Bars
Search for setting


Attrition vs Job Satisfaction
by Rajendra Kumar (B24BS2273)
Create a story
Export & publish
968
auto
Preview
Data
Saved
4
2
3
1
Number of Employees Leaving
4
3
2
1
52
73
46
66
0 10 20 30 40 50 60 70 80
Job Satisfaction Level (1–4)
Help
Line, bar and pie charts
v39.0.0
Theme
No theme
Chart type
Bar chart (grouped)
Grid mode
Single chart
Grid of charts
Height mode
Auto
Standard
Aspect ratio
Aggregation mode
None
Sum
Average
Count
Controls & filters
Colors
Bars
Labels
Search for setting

11. Napkin AI Insights

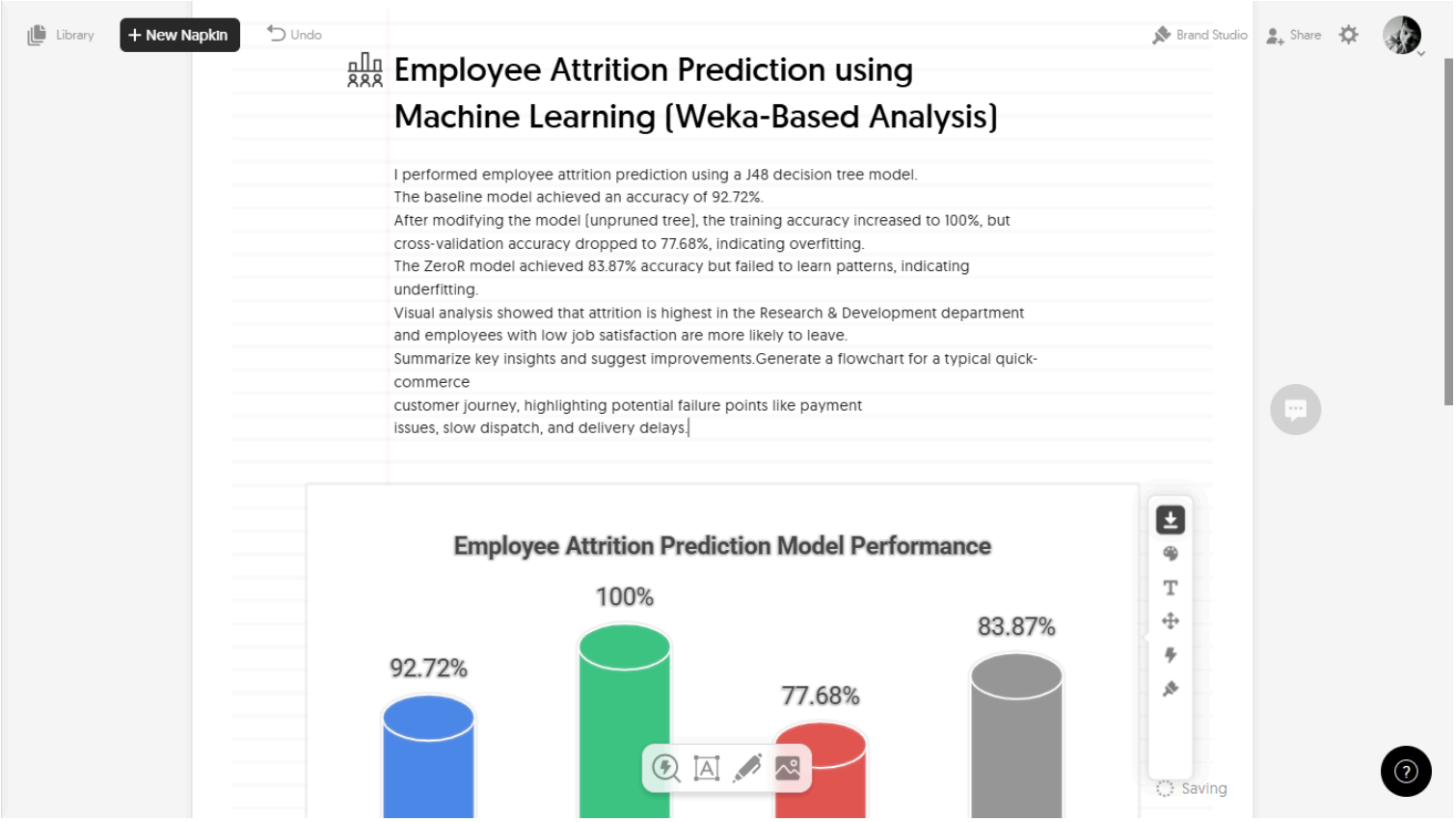
Napkin AI was used to summarize the results of the machine learning models.

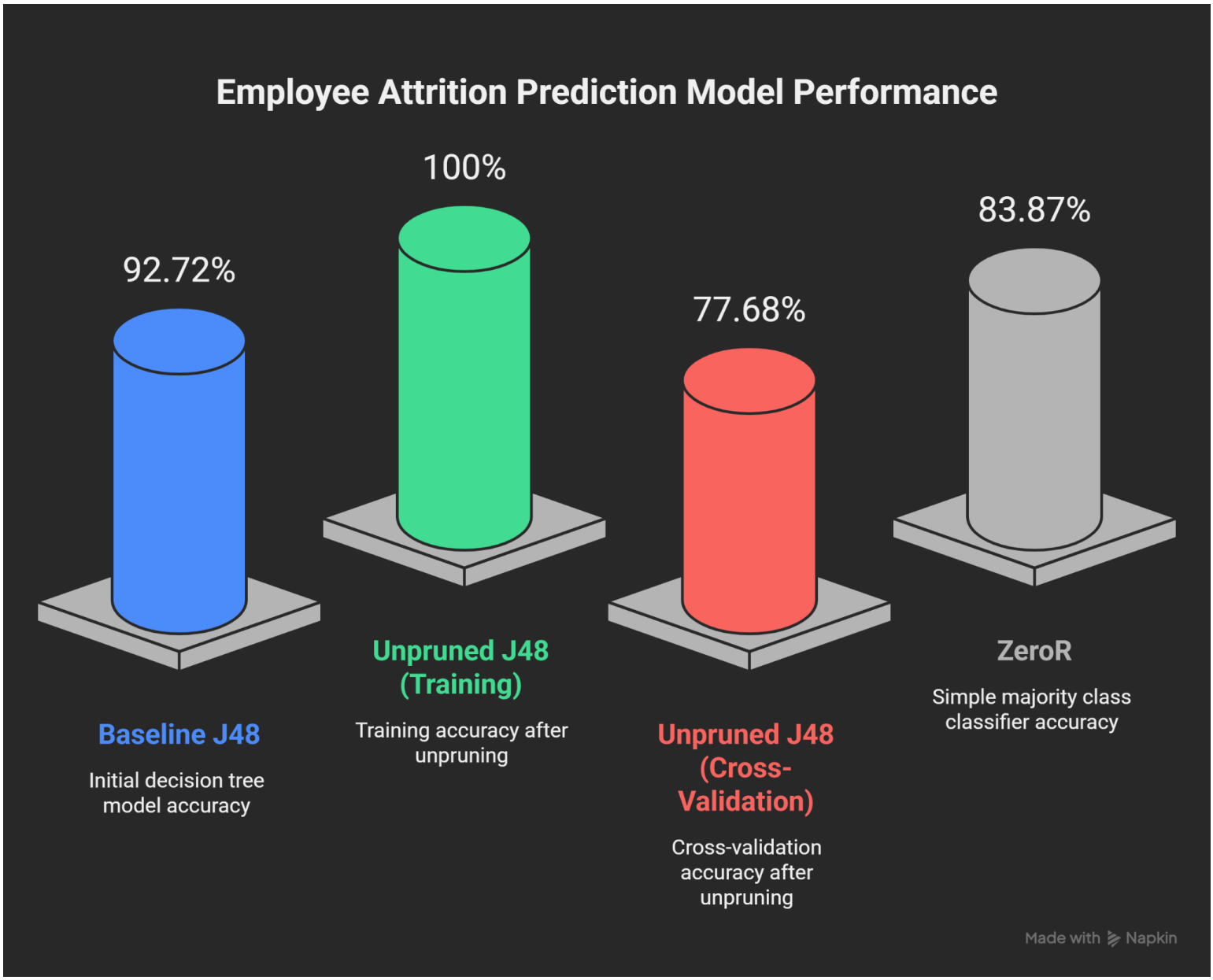
The outputs from different models were given as input, and it generated key insights.

It clearly showed overfitting and underfitting and explained the model performance in a simple way.

It also suggested some improvements to make the model better.

This step helped in understanding the results more clearly.




Employee Attrition Prediction Model Performance
92.72%
100%
77.68%
83.87%
Baseline J48
Initial decision tree model accuracy
Unpruned J48 (Training)
Training accuracy after unpruning
Unpruned J48 (Cross-Validation)
Cross-validation accuracy after unpruning
ZeroR
Simple majority class classifier accuracy
Made with Napkin

🟣 12. Reflection Answers

1. Important attributes:

JobSatisfaction, OverTime, MonthlyIncome, YearsAtCompany

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2. Overfitting or Underfitting:

The unpruned J48 model showed overfitting because training accuracy was very high while test accuracy dropped.

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3. Data improvements:

- Add more features (work environment, manager feedback)
- Remove irrelevant attributes
- Balance dataset

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4. HR usage:

HR teams can identify high-risk employees and take actions like promotions, workload reduction, and training.

⭐ ⭐ Thanks ⭐ ⭐

Submitted by: Rajendra Kumar(B24Bs2273)